

MYELOID NEOPLASIA

FAP-1 loss impairs megakaryocyte demarcation membrane system and platelet function with myelofibrosis-like features

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KEY POINTS

- FAP-1 loss disrupts DMS formation in MKs, which impairs maturation and platelet function and predisposes to MF-like features with aging.
- FAP-1 deficiency sustains Src activation, blocks cofilin-mediated actin disassembly, and arrests MK at the pre-DMS stage.

Fas-associated phosphatase 1 (FAP-1), a nonreceptor protein tyrosine phosphatase, has been implicated in multiple signaling pathways, but its *in vivo* role remains unclear. In this study, we showed that FAP-1-deficient (FAP-1^{ΔP/ΔP}) mice develop early megakaryocyte (MK) hyperplasia with defective platelet function and occasional hemorrhagic manifestations, accompanied by myelofibrosis (MF)-like features, in aged animals. Bone marrow analyses revealed impaired demarcation membrane system (DMS) development with pre-DMS arrest in MK, which led to defective proplatelet formation, impaired platelet function, and prolonged bleeding that was partly associated with reduced clot retraction. Mechanistically, a FAP-1 deficiency induces sustained Src activation and cofilin inactivation, thereby impairing the perinuclear actin remodeling required for DMS expansion and leading to pre-DMS arrest. With aging, approximately half of the mice developed a symptomatic phenotype, characterized by extramedullary hematopoiesis, hepatosplenomegaly, anemia, and thrombocytopenia; among these, most remained in a prefibrotic state, whereas a subset progressed to fibrosis-like changes. A bone marrow

transplantation demonstrated that the MK abnormalities and fibrosis-associated changes were hematopoietic cell-intrinsic and partially transferable. Pharmacologic inhibition of Src with dasatinib attenuated these defects in FAP-1-deficient mice, supporting pathway specificity. In patients with primary MF, reduced FAP-1 expression is associated with pre-DMS MK accumulation, abnormal DMS and actin organization, and Src activation, thus supporting clinical relevance. Collectively, we identified a FAP-1-dependent mechanism that governs the Src-cofilin-mediated actin remodeling required for MK maturation and platelet function and suggest this pathway as a potential therapeutic target for platelet dysfunction, hemorrhagic complications, and fibrosis-associated disease.

Introduction

Fas-associated phosphatase 1 (FAP-1) is a large, non-transmembrane protein tyrosine phosphatase that functions as a scaffold and contains an N-terminal kinase non-catalytic C-lobe (KIND) domain, a 4.1 protein, ezrin, radixin, moesin (FERM) domain, 5 PSD95/Dlg/ZO-1 (PDZ) domains, and a C-terminal catalytic protein tyrosine phosphatase (PTP) domain.¹ The FERM domain mediates submembranous localization and links FAP-1 to the actin cytoskeleton,¹ whereas the PDZ domains enable interactions with diverse proteins. Through its catalytic activity, FAP-1 dephosphorylates multiple substrates, including Src, thereby modulating diverse biologic

processes.¹ However, its precise function and relevance to human disease remain unclear. Reduced FAP-1 expression in several cancers suggests that it plays a tumor-suppressive role, whereas in other malignancies, it seems to promote tumor growth, thus highlighting context-dependent functions.¹⁻³

Although FAP-1 has been implicated in the regulation of apoptosis, cytokinesis, and cell-cell junctions *in vitro*,¹ these functions remain unverified *in vivo*. Moreover, FAP-1-deficient mice show no spontaneous tumors or overt morphological abnormalities and display only modest phenotypes under stress.⁴⁻⁶ However, during a 2-year follow-up, we observed that FAP-1 enzyme-deficient (FAP-1^{ΔP/ΔP}) mice developed

late-onset primary myelofibrosis (PMF)-like features, with some mice dying with prominent intra-abdominal hemorrhage, revealing an unexpected association between FAP-1 deficiency, megakaryocyte (MK) or platelet dysfunction, and PMF-like manifestations with bleeding complications.

PMF is the most aggressive myeloproliferative neoplasm with a higher risk for progression to acute myeloid leukemia than polycythemia vera or essential thrombocythemia (ET).^{7,8} Although ~90% of PMF and ET cases harbor JAK2, MPL, or CALR mutations that activate MPL/JAK/STAT signaling and drive MK expansion,^{8,9} these diseases diverge at the level of MK maturation and function. In ET, MKs are relatively mature and produce functional platelets, which leads to thrombocytosis and thrombosis as the predominant complications.^{10,11} In contrast, PMF is characterized, even at the prefibrotic stage, by dysplastic, immature MKs with cloud-like or bulbous nuclei, hyperchromasia, and tight clustering,¹¹ which are associated with defective proplatelet formation, abnormal platelets, and a bleeding-prone phenotype.^{12,13}

Aberrant MKs also play a central role in bone marrow fibrosis by promoting fibroblast proliferation through the secretion of cytokines, such as transforming growth factor- β (TGF- β), vascular endothelial growth factor, and platelet-derived growth factor.¹⁴ Clinically, prefibrotic PMF (pre-PMF) progresses to overt PMF in an age-dependent manner, with ~10% to 15% of patients transforming within 10 years and ~35% to 40% transforming by 15 years, whereas ~70% remain prefibrotic at 10 years.^{15,16} Extramedullary hematopoiesis (EMH), a hallmark of PMF, is already present in pre-PMF and involves the spleen and liver, which leads to hepatosplenomegaly with multilineage hematopoiesis, and occurs more frequently in overt disease.^{17,18} Together, these observations support the current models that place MK pathology at the apex of PMF pathogenesis, with EMH, hepatosplenomegaly, and fibrosis arising as downstream consequences during disease evolution.^{19,20} Accordingly, defining mechanisms that impair MK maturation and platelet function may provide insight into PMF pathogenesis and inform strategies to limit fibrotic progression and the associated hemorrhagic complications.

MKs are specialized bone marrow cells that undergo terminal thrombopoietic maturation.^{21,22} Under thrombopoietin (TPO) stimulation, MK progenitors undergo endomitosis to become polyploid, synthesize granule proteins, and form the demarcation membrane system (DMS), which is remodeled into proplatelets that extend into sinusoidal vessels to release platelets.²³ Transmission electron microscopy (TEM) analyses showed that MKs in pre-PMF and overt PMF exhibit marked DMS reduction, consistent with arrest at a pre-DMS stage across fibrosis grades, suggesting an early and fundamental MK maturation defect that precedes overt fibrosis.²⁴ Because DMS disorganization may impair proplatelet formation and generate dysfunctional platelets, it may contribute to thrombocytopenia, bleeding, and marrow fibrosis.²⁴⁻²⁶ These findings implicate defective DMS development in MK maturation defects in PMF and highlight DMS disruption as a key pathogenic mechanism.

In our FAP-1 ^{$\Delta P/\Delta P$} model, loss of FAP-1 sustains Src activation and arrests MKs at the pre-DMS stage from an early age,

which leads to highly penetrant platelet dysfunction and age-dependent emergence of myelofibrosis (MF)-like features in a subset of mice. Together, these findings indicate that FAP-1 is a regulator of MK maturation through DMS formation and suggest that dysregulated FAP-1–Src signaling drives early platelet dysfunction and is associated with the development of MF-like features with aging, with potential relevance to PMF pathogenesis and the associated bleeding complications.

Methods

A detailed description of all the experimental procedures is available in the supplemental information on the *Blood* website.

FAP-1-deficient mouse model

FAP-1 ^{$\Delta P/\Delta P$} mice on a C57BL/6N $\times$ C57BL/6J background were generated by crossing FAP-1 ^{$+/ \Delta P$} C57BL/6N mice (originally generated by Hendriks et al⁴) with C57BL/6J wild-type (WT) mice and interbreeding the offspring. Age-matched WT controls were obtained by crossing the C57BL/6N and C57BL/6J strains.

BMT

Bone marrow transplantations (BMTs) were performed as previously described.²⁷ Recipient mice received lethal total-body irradiation, followed by IV injection of donor bone marrow cells on the same day. The mice were monitored for hematologic recovery and morbidity.

TEM

Femurs and tibias were processed for TEM at the Microscope Core Laboratory, Chang Gung Memorial Hospital. Ultrathin bone marrow sections were examined using a Hitachi HT7800 TEM.

Cell lines

HEL and M-07e cells (DSMZ, ACC-104) were cultured in RPMI 1640 with fetal bovine serum, L-glutamine, and penicillin-streptomycin; granulocyte-macrophage colony-stimulating factor (10 ng/mL) was added for the M-07e cells.

G-actin and F-actin fractionation

G-actin and F-actin fractionations were performed using a commercial kit (Cytoskeleton, Inc), according to the manufacturer's protocol, with minor modifications, and were analyzed by immunoblotting.

In vitro MK differentiation assay

Hematopoietic progenitor cells were isolated from femurs and tibias²⁸ and enriched by magnetic separation. The cells were cultured in serum-free medium containing stem cell factor and TPO.

Proplatelet formation assay

Proplatelet formation was assessed using mature bone marrow MKs purified by bovine serum albumin gradient sedimentation and plated on fibrinogen-coated coverslips. The cells were incubated for 16 to 18 hours at 37°C before analysis.

Tail bleeding assay

Tail bleeding assays were performed as described previously²⁹ using cumulative bleeding time as the primary end point.

Clot retraction assay

Clot retraction was assessed in platelet-rich plasma preparations upon thrombin stimulation, and the extent of retraction was quantified.

Assessment of platelet activation by surface P-selectin expression

Platelet activation was evaluated by flow cytometric measurement of surface P-selectin expression on CD41-positive platelets following stimulation with adenosine diphosphate or thrombin.

Clinical samples

Formalin-fixed paraffin-embedded bone marrow tissues from patients with ET, PMF, and lymphoma were obtained from the Department of Pathology, National Taiwan University Hospital. Histologic features were evaluated on these sections by immunofluorescence (IF), immunohistochemistry, and reticulin staining.

Results

Aging FAP-1^{ΔP/ΔP} mice developed hepatosplenomegaly with EMH caused by MF

FAP-1^{ΔP/ΔP} mice on a C57BL/6N × C57BL/6J hybrid background exhibited progressive weight loss and accelerated mortality beginning at ~500 days (Figure 1A). Subsequent necropsies demonstrated hepatosplenomegaly in most mice (Figure 1B), alongside a spontaneous hemoperitoneum in 50% of the mice, neither of which occurred in age-matched WT controls.

To investigate the cause of death, FAP-1^{ΔP/ΔP} mice that lost >20% of their body weight within a month underwent pathologic examinations, which consistently revealed hepatosplenomegaly with disrupted architecture and bone marrow-like hematopoietic infiltration, indicative of EMH (supplemental Figure 1A-D). Hemograms showed anemia and severe thrombocytopenia, and the bone marrow histology demonstrated atypical MK expansion with fibrosis and occasional osteosclerosis (supplemental Figure 1E-G).

To validate these observations, 64 FAP-1^{ΔP/ΔP} mice were followed for up to 26 months, and 32 mice developed symptomatic disease, characterized by hepatosplenomegaly with EMH composed of multilineage hematopoietic cells with minimal lymphoid components (Figure 1C-F; supplemental Figure 2A-H; symptomatic). These mice also exhibited severe anemia and thrombocytopenia with teardrop-shaped red blood cells, a hallmark of MF³⁰ (Figure 1G-I; supplemental Table 1; symptomatic). Bone marrow examinations confirmed atypical MK accumulation in all symptomatic mice with prefibrotic changes (31.3%), MF (59.4%), and osteosclerosis (9.3%) (Figure 1J; supplemental Figure 3D-E; symptomatic). The remaining 32 mice were asymptomatic, showing only mild thrombocytopenia with prefibrotic, atypical MK accumulation (Figure 1C-D, 1G-J; supplemental Figures 2A and 3D-E; supplemental Table 1; asymptomatic). Notably, these disease

features were independent of sex (supplemental Figure 4). Bone marrow analyses revealed stage-dependent changes in cellularity with increased cellularity in early stages (MF grade 0/1) and decreased cellularity in advanced fibrosis (MF grade 2/3), accompanied by persistently increased microvessel density and reduced Ter-119 staining, consistent with a progression from prefibrotic to high-grade MF-like features (supplemental Figure 3).

Collectively, these data show that a FAP-1 deficiency may induce MK accumulation and MF-like changes, accompanied by EMH, anemia, thrombocytopenia, and spontaneous hemorrhage, which closely resemble clinical PMF with bleeding complications.

Early MK abnormalities precede an age-dependent, hematopoietic cell-intrinsic MF-like phenotype in FAP-1^{ΔP/ΔP} mice

To define the temporal relationship between MK abnormalities and MF, we analyzed mice at different ages with stratification into asymptomatic or symptomatic. The bone marrow MKs were increased by 2 to 4 months and remained elevated thereafter in both asymptomatic and symptomatic FAP-1^{ΔP/ΔP} mice (Figure 2A). Despite this, the platelet counts in FAP-1^{ΔP/ΔP} mice were initially comparable with that of WT mice but failed to increase with age, leading to relative thrombocytopenia with aging, which became severe in symptomatic mice (Figure 2B). The serum TPO levels were elevated in young FAP-1^{ΔP/ΔP} mice and remained higher in asymptomatic mice, whereas WT mice showed an age-related increase in platelet counts with reduced TPO, consistent with feedback regulation³¹ (Figure 2C). In symptomatic mice, severe thrombocytopenia coincided with reduced TPO (Figure 2A and 2C), consistent with impaired hepatic TPO production caused by liver dysfunction associated with hepatic EMH.³² Notably, overt MF phenotypes, including anemia, thrombocytopenia, hepatosplenomegaly, and bone marrow fibrosis, emerged progressively with age, starting with a single mouse at 10 to 15 months, increasing to 4 mice (~21%) at 16 to 21 months, and 27 mice (~46%) at 22 to 26 months (Figure 2A-F; supplemental Figure 5; supplemental Table 2).

Given the established role of MK-derived TGF-β1 and P-selectin in MF pathogenesis,^{9,33} we subsequently assessed TGF-β1 and P-selectin in FAP-1^{ΔP/ΔP} mice. The serum TGF-β1 levels were elevated in 22- to 26-month-old FAP-1^{ΔP/ΔP} mice and were accompanied by increased marrow TGF-β1 expression and a higher proportion of TGF-β1-expressing MKs (Figure 2G-I), suggesting that the expansion of TGF-β1-producing MKs may promote the age-dependent transition from MK hyperplasia to progressive marrow fibrosis. In line with disease progression, activated P-selectin-high MKs were elevated in both asymptomatic and symptomatic mice with the highest levels measured in symptomatic mice (supplemental Figure 6). Together, these data indicate that megakaryocytic expansion and TPO dysregulation precede overt MF in FAP-1^{ΔP/ΔP} mice in the absence of antecedent thrombocythemia or erythrocytosis, supporting a primary MF-like process.

To assess whether the MK abnormalities and the age-dependent MF-like phenotype were driven by a hematopoietic cell-intrinsic mechanism or the aging bone marrow microenvironment, we performed reciprocal BMTs. FAP-1^{ΔP/ΔP}

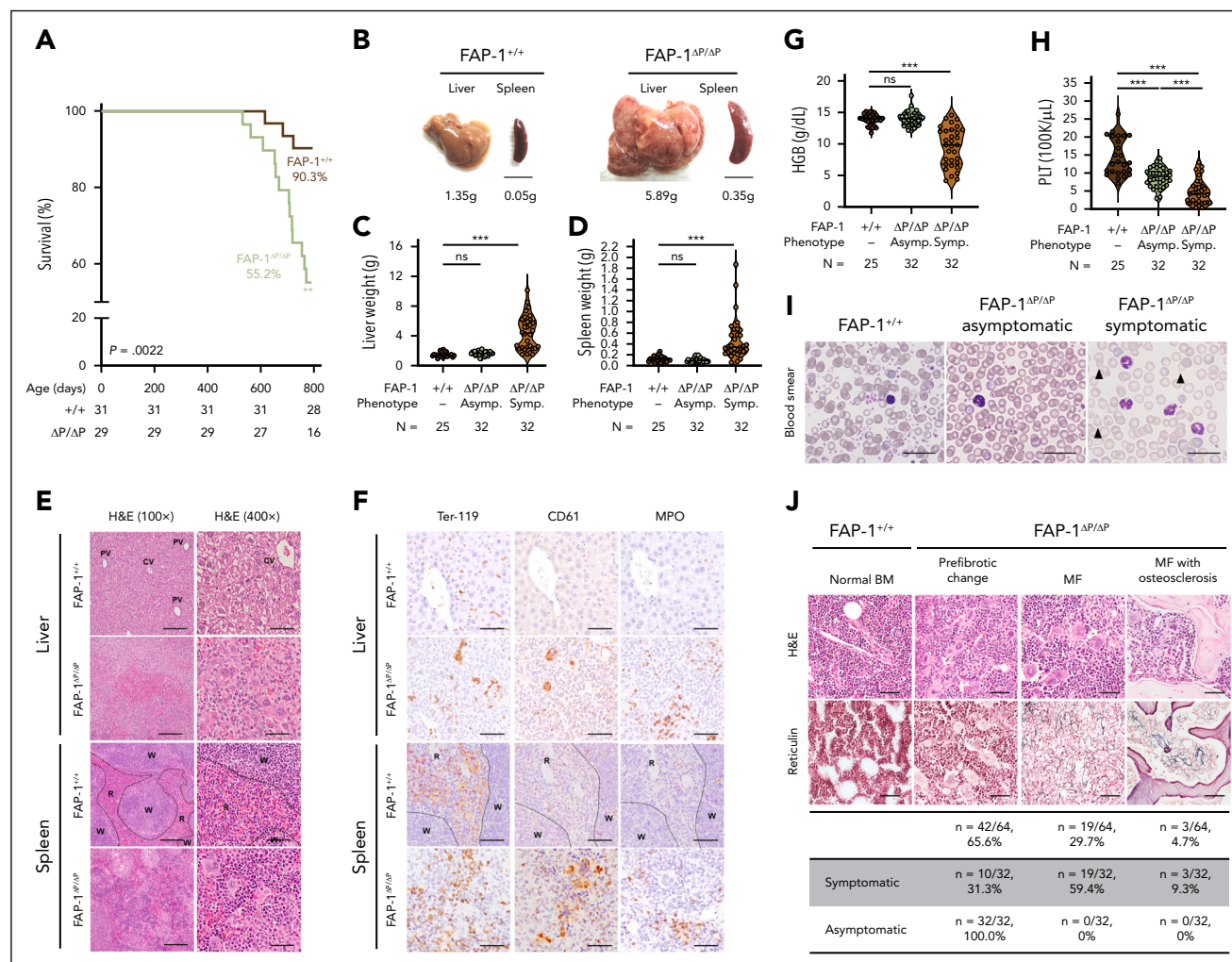


Figure 1. Aged FAP-1^{ΔP/ΔP} mice (22 to 26 months) developed extramedullary hematopoiesis and progressive MF-like phenotypes. (A) Kaplan-Meier survival curves of FAP-1^{+/+} (N = 31) and FAP-1^{ΔP/ΔP} (N = 29) mice. The statistics were calculated using the χ^2 test. (B) Representative gross morphology of the livers and spleens from healthy FAP-1^{+/+} and deceased FAP-1^{ΔP/ΔP} mice. The organ weights (g) are shown below each image. The scale bars indicate 1 cm. Quantification of the liver (C) and spleen weights (D) in FAP-1^{+/+} and FAP-1^{ΔP/ΔP} mice. The FAP-1^{ΔP/ΔP} mice were stratified into symptomatic (symp.) and asymptomatic (asympt.) groups based on phenotype. (E) Representative hematoxylin and eosin (H&E) staining of the livers and spleens from FAP-1^{+/+} and symptomatic FAP-1^{ΔP/ΔP} mice. The right panels provide higher-magnification views of regions depicted in the left panels. In FAP-1^{+/+} livers, PV and CV denote the portal vein and central vein, respectively. In FAP-1^{+/+} spleens, red pulp and white pulp are delineated by dotted lines and are labeled as "R" and "W", respectively. The scale bars indicate 200 μ m (100 $\times$) and 50 μ m (400 $\times$). (F) Representative immunohistochemistry staining of livers and spleens from FAP-1^{+/+} and symptomatic FAP-1^{ΔP/ΔP} mice. Ter-119 was used to identify the erythroid lineage, CD61 was used for the MK lineage, and myeloperoxidase (MPO) was used for the myeloid lineage. The scale bars indicate 50 μ m. (G-H) Peripheral blood analysis of the hemoglobin (HGB) levels (G) and platelet counts (PLT) (H) in the FAP-1^{+/+} and FAP-1^{ΔP/ΔP} mice. (I) Representative peripheral blood smears from FAP-1^{+/+} and FAP-1^{ΔP/ΔP} mice. The arrowheads highlight teardrop-shaped red blood cells in symptomatic FAP-1^{ΔP/ΔP} mice. The scale bars indicate 20 μ m. (J) Representative H&E and reticulin staining of bone marrow from FAP-1^{+/+} and FAP-1^{ΔP/ΔP} mice. The scale bars indicate 50 μ m. A summary table below indicates the frequencies of the various bone marrow phenotypes observed in FAP-1^{ΔP/ΔP} mice. Data in panels C, D, G, and H are presented as violin plots. The solid lines denote the median, and the dashed lines indicate the quartiles. All statistical analyses were performed using t tests. * $P < .05$, ** $P < .01$, *** $P < .001$; ns, not significant.

marrow from symptomatic donors (low grade MF 0/1) induced early mortality, hepatosplenomegaly, and MK hyperplasia in WT recipients, whereas WT marrow rescued the survival and hematopoiesis in FAP-1^{ΔP/ΔP} recipients (supplemental Figure 7). Moreover, only marrow from symptomatic donors, but not that from young or asymptomatic donors, transmitted disease to young WT recipients, including hepatosplenomegaly, thrombocytopenia, early mortality (2/4 by day 20), and histologic MF in 1 survivor by day 60 (supplemental Figure 8). These findings demonstrate that the MK abnormalities and MF-like features in FAP-1^{ΔP/ΔP} mice are hematopoietic cell-intrinsic.

FAP-1 deficiency impairs DMS maturation, proplatelet formation, and platelet function

Thrombocytopenia, despite MK hyperplasia, indicates defective MK maturation; we therefore assessed MK differentiation at 22 to 26 months when this discordance was maximal. Flow cytometry analyses showed predominantly 8N to 16N MKs in aged WT and asymptomatic FAP-1^{ΔP/ΔP} mice with a shift toward 2N to 4N MKs in symptomatic mice, whereas platelet factor-4 staining was similar across the groups (Figure 3A-B). TEM revealed progressive DMS disruption. In WT mice, >90% of MKs exhibited continuous, highly branched DMS. In asymptomatic FAP-1^{ΔP/ΔP} mice, MKs showed partial

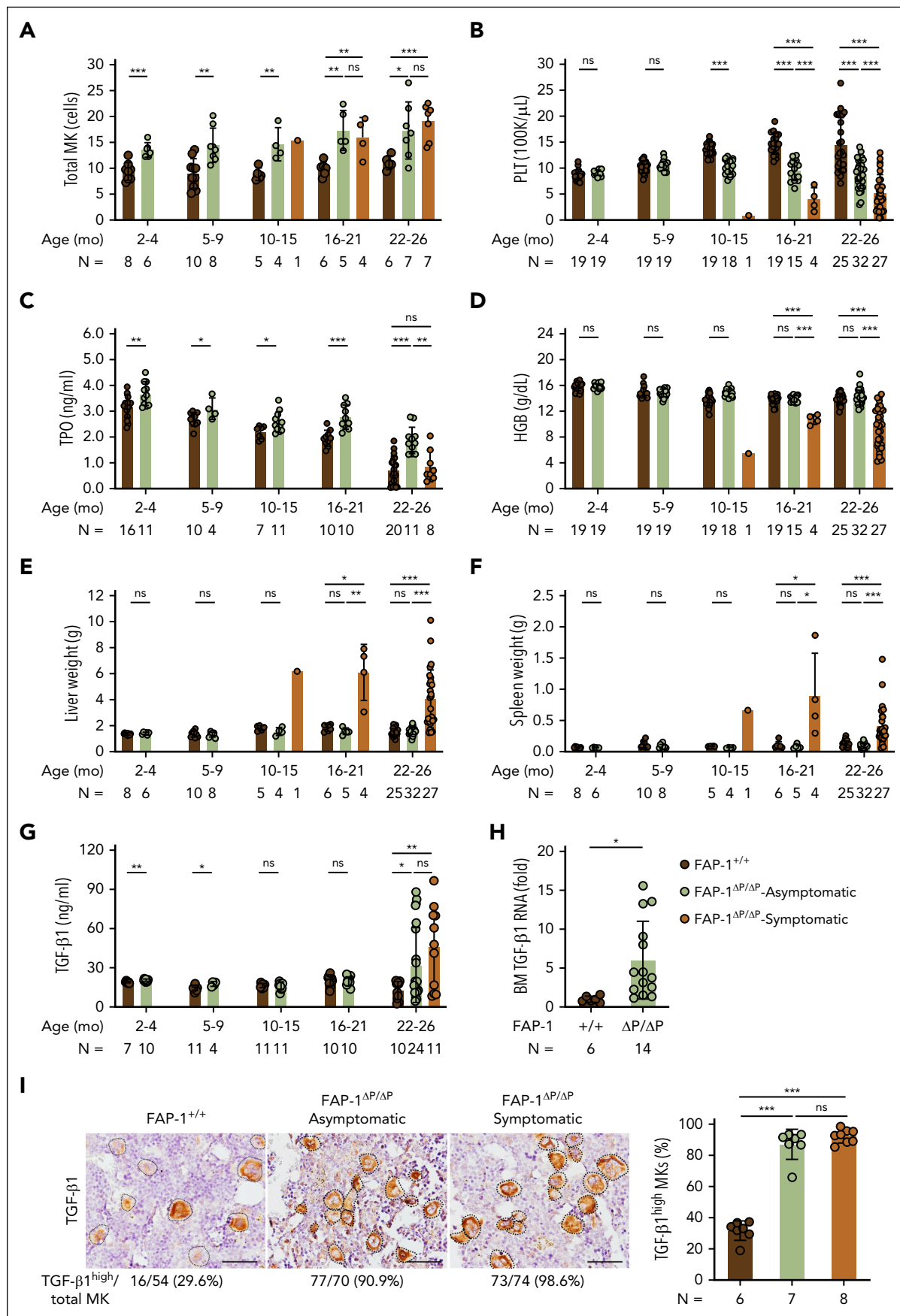


Figure 2.

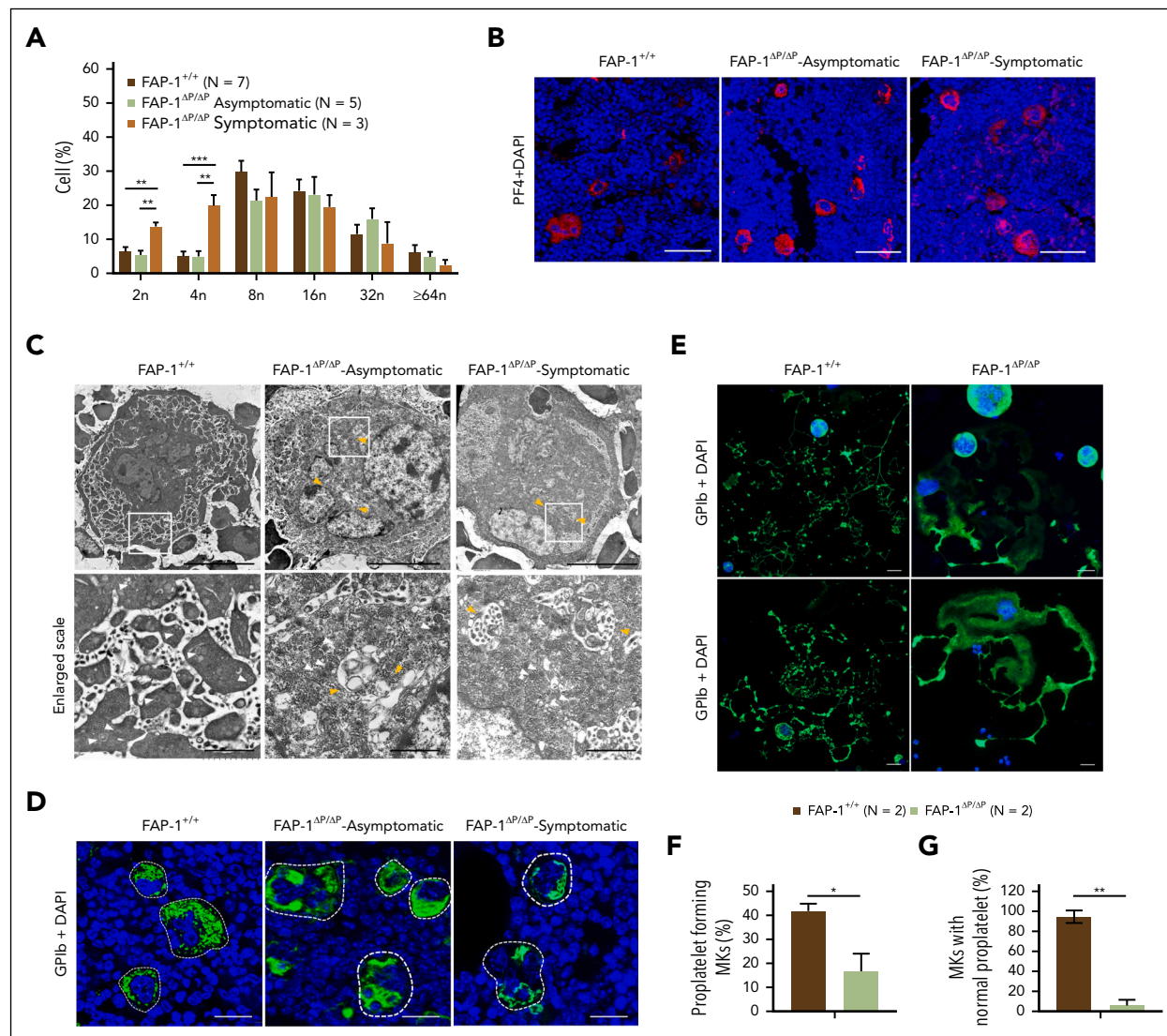


Figure 3. FAP-1 deficiency blocks MK differentiation by impairing DMS development. (A) Flow cytometry analysis of the DNA content (ploidy) in bone marrow cells from aged FAP-1^{+/+}, asymptomatic, and symptomatic FAP-1^{ΔP/ΔP} mice. The sample sizes (N) are indicated in the graph. (B) Representative IF staining images of platelet factor-4 (PF4; red) in bone marrow sections. The nuclei were counterstained with DAPI (4',6-diamidino-2-phenylindole; blue). The scale bars indicate 50 μ m. (C) Representative TEM analyses of bone marrow MKs (N = 3 mice per group). Approximately 10 to 15 MKs were analyzed per mouse; additional representative images are shown in supplemental Figure 9A. The white arrowheads indicate granules, and the yellow arrowheads indicate disorganized pre-DMS structures. The regions shown at enlarged scale are indicated by the white squares in the upper panels. The scale bars indicate 10 μ m (upper panel) and 2 μ m (lower panel). (D) Representative IF staining images of GPIb (green) in bone marrow sections (N = 5 mice per group). The white dashed lines outline individual MKs. Additional representative images from other mice are shown in supplemental Figure 9B. The scale bars indicate 20 μ m. (E) Representative IF images of proplatelet formation in cultured MK derived from aged (22- to 26-month-old) mice (N = 2-3 per group). The cells were stained for GPIb (green) and nuclei (blue). The scale bars indicate 20 μ m. (F-G) Quantification of the percentage of proplatelet-forming MKs (F) and MKs with normal proplatelet structure (G). A total of 50 cells were analyzed for each mouse. Data in panels A, F, and G are presented as mean $\pm$ standard deviation. Statistical significance was determined using a t test. **P* < .05; ***P* < .01; ****P* < .001.

disorganization with ~40% retaining some intact DMS. In symptomatic FAP-1^{ΔP/ΔP} mice, MKs displayed severe fragmentation with loss of membrane continuity, leading to poorly

defined platelet territories in >90% of cells (Figure 3C; supplemental Figure 9A). Consistently, glycoprotein Ib (GPIb) staining indicated perinuclear pre-DMS arrest,³⁴ which was

Figure 2. FAP-1 deficiency causes early megakaryocytic and TPO abnormalities and age-dependent penetrance of a PMF-like phenotype. Temporal analysis of the total bone marrow MK counts (A), PLT counts (B), serum TPO levels (C), HGB levels (D), liver weight (E), spleen weight (F), and serum TGF- β 1 levels (G) in FAP-1^{+/+} and FAP-1^{ΔP/ΔP} mice aged 2 to 26 months. (H) Reverse transcriptase-quantitative polymerase chain reaction analysis of TGF- β 1 mRNA expression in whole bone marrow cells from aged mice (22-26 months). (I) Left panel: Representative immunohistochemistry staining of TGF- β 1 in bone marrow sections from 22- to 26-month-old FAP-1^{+/+}, asymptomatic FAP-1^{ΔP/ΔP}, and symptomatic FAP-1^{ΔP/ΔP} mice. Individual MKs are outlined with dotted lines. Right panel: The bar graph quantifies the percentage of TGF- β 1^{high} MKs. The numbers below each image indicate the count of TGF- β 1^{high} MKs and the total number of MKs for the representative mouse shown. The scale bars indicate 50 μ m. Data are presented as the mean $\pm$ standard deviation. The sample sizes (N) are indicated below the x-axis for each group. Statistical significance was determined using a t test. **P* < .05, ***P* < .01, ****P* < .001; ns, not significant. For ease of comparison, data for the 22- to 26-month cohort in panels (A-D) were replotted from Figure 1C, Figure 1D, Figure 1G, and Figure 1H, respectively.

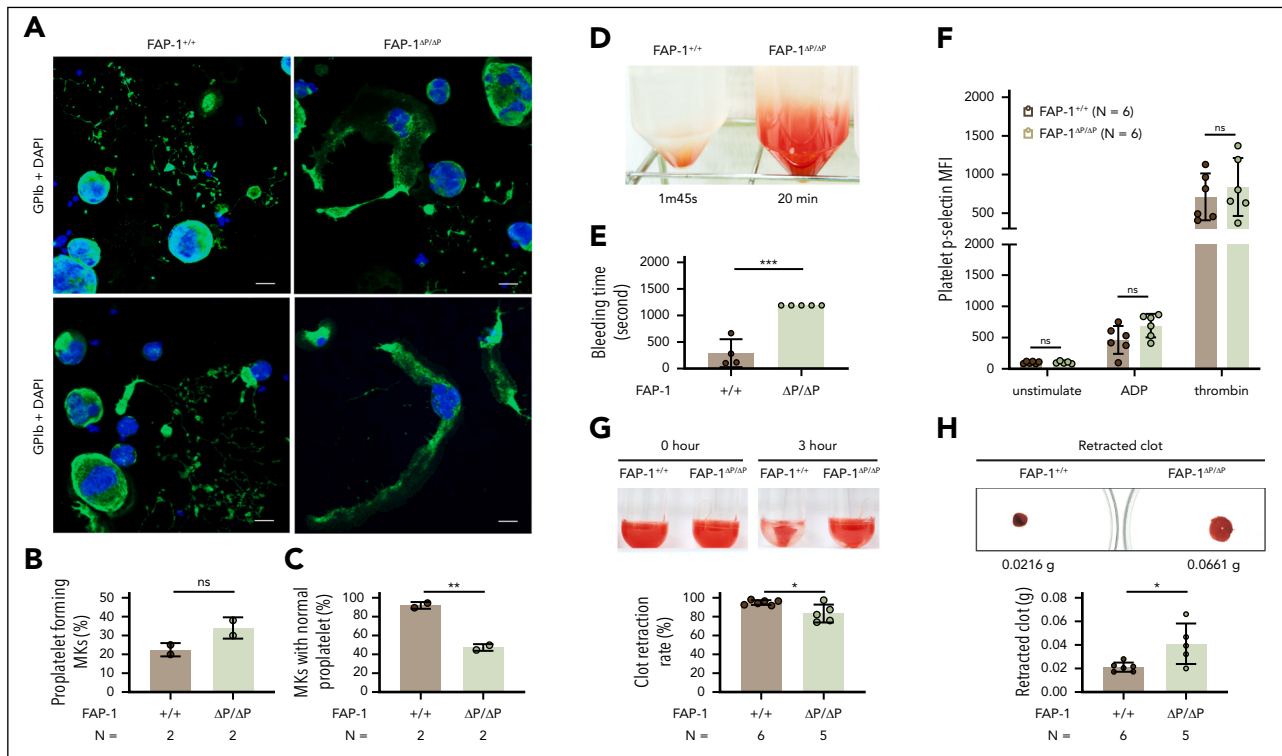


Figure 4. FAP-1 deficiency leads to aberrant proplatelet formation and impaired platelet function in 2- to 4-month-old mice. (A) Representative IF images of proplatelet formation in cultured MKs from young (2- to 4-month-old) mice (N = 2 per group). The cells were stained for GPIb (green) and nuclei (blue). The scale bars indicate 20 μ m. (B-C) Quantification of the percentage of proplatelet-forming MKs (B) and MKs with normal proplatelet structure (C). A total of 50 cells were analyzed for each mouse. (D-E) In vivo tail bleeding assay. A 3 to 5 mm segment of the tail tip was excised from FAP-1^{+/+} and FAP-1 Δ P/ Δ P mice, and the total blood loss (D) and bleeding time (E) were measured in 37°C saline. (F) Flow cytometric analysis of P-selectin surface expression (marker of alpha-granule) on platelets following stimulation with adenosine diphosphate (ADP; 0.1mM) or thrombin (0.1 U/mL). (G) Upper panel: representative photographs of clot retraction assays performed with platelet-rich plasma at 0 and 3 hours. Lower panel: quantification of the clot retraction rate (%). The clot retraction rate (%) was calculated as (residual volume / total reaction volume) $\times$ 100. (H) Upper panel: representative image of retracted clots. Lower panel: quantification of the retracted clot weight. The sample size (N) in each group is listed under the graphs. The data in panels B, C, E, and F and lower panel of G and H are presented as the mean $\pm$ standard deviation *P < .05; **P < .01; ***P < .001.

more pronounced in symptomatic MKs (Figure 3D; supplemental Figure 9B). Functionally, ~50% of WT MKs formed long, branched proplatelets, whereas most FAP-1-deficient MKs failed to form typical proplatelets, with <15% retaining short, poorly branched protrusions with fewer swellings (Figure 3E-G).

Consistently, these defects were recapitulated in vitro and partially rescued by re-expression of WT FAP-1, supporting a cell-intrinsic role for FAP-1 (supplemental Figure 10). A BMT from symptomatic FAP-1 Δ P/ Δ P marrow donors reproduced severe DMS abnormalities and thrombocytopenia in WT recipients (supplemental Figures 7J and 8I), supporting a hematopoietic cell-intrinsic defect.

To define the onset of DMS abnormalities, GPIb staining revealed pre-DMS features in <10% of WT MKs vs in ~25% of FAP-1 Δ P/ Δ P MKs at 2 to 4 months, which increased to >40% with age (supplemental Figure 11A-B), indicating progressive DMS disruption. Consistently, proplatelet analysis demonstrated abnormal proplatelet morphology in ~40% of MKs from 2- to 4-month-old FAP-1 Δ P/ Δ P mice (Figure 4A-C), which rose to >90% in aged mice (Figure 3E-G). This progressive impairment in proplatelet formation likely contributed to age-dependent thrombocytopenia despite MK hyperplasia.

Because platelets arise from DMS-dependent proplatelet extensions, we assessed platelet function in 2- to 4-month-old FAP-1 Δ P/ Δ P mice with platelet counts that were comparable with WT mice despite early DMS abnormalities. The FAP-1 Δ P/ Δ P mice exhibited increased blood loss and prolonged bleeding times, indicating platelet dysfunction (Figure 4D-E). Granule secretion was preserved with agonist-induced surface P-selectin being comparable with WT mice (Figure 4F), whereas clot retraction was impaired with reduced serum extrusion and increased clot weight (Figure 4G-H).

Together, these findings indicate that a FAP-1 deficiency was associated with progressive defects in DMS maturation, proplatelet formation, and platelet hemostatic function, which likely underlie the age-dependent thrombocytopenia and bleeding tendency that precede the development of MF-like phenotypes in this model.

FAP-1-dependent actin depolymerization contributes to DMS expansion during MK maturation

DMS biogenesis involves 3 actin-regulated steps, namely focal membrane assembly at the GPIb-V-IX-enriched periphery; plasma membrane invagination that forms perinuclear

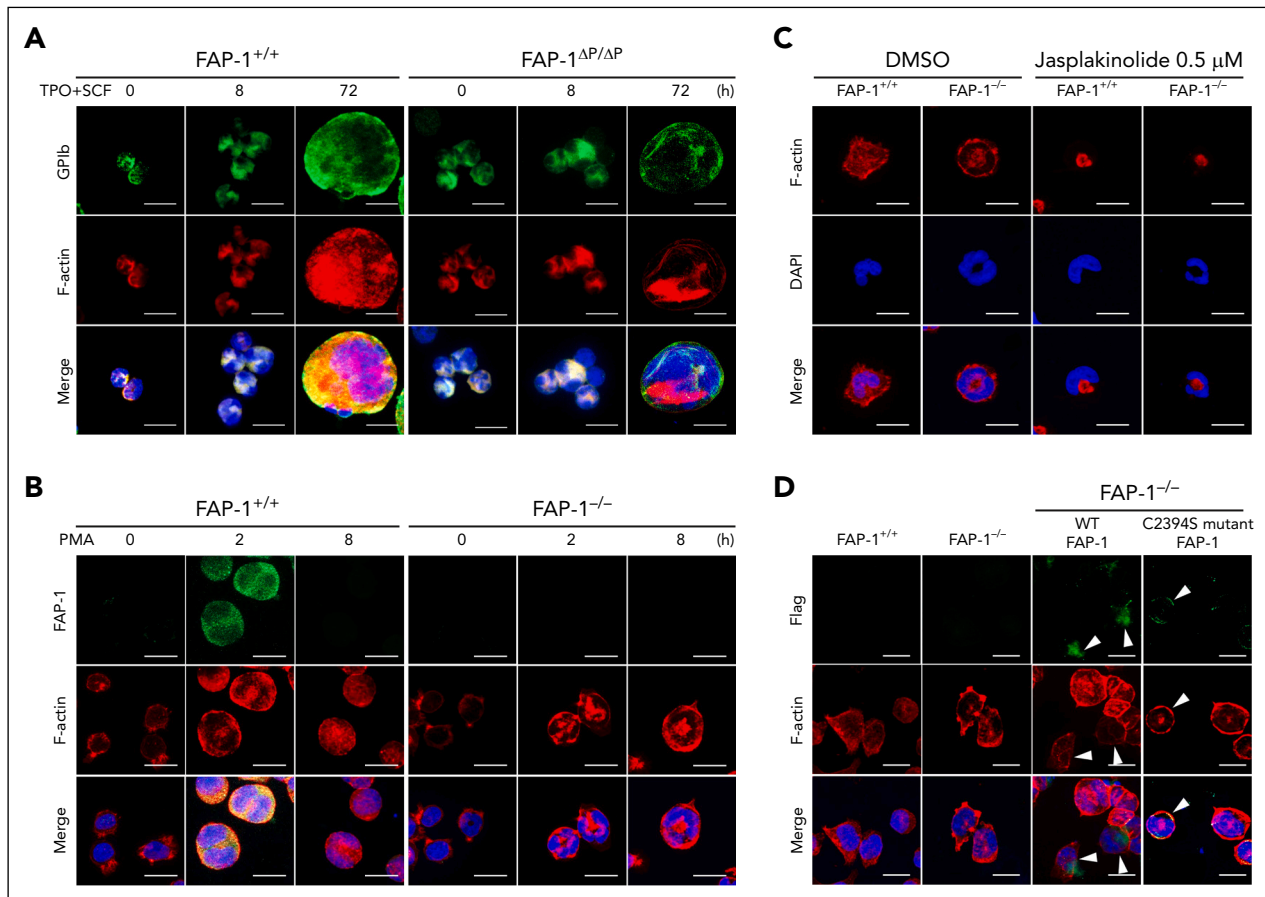


Figure 5. FAP-1 enzyme activity-dependent perinuclear actin depolymerization is essential for DMS expansion during MK differentiation. (A) IF staining of GPIb (green) and F-actin (red) in cultured MKs stimulated with TPO and stem cell factor for the indicated times. A total of 5 to 8 MKs per mouse from 3 independent mice per group (>15 MKs total per group) were analyzed. The reported phenotype was present in the majority of the analyzed MKs (>70%) across all mice. The scale bars indicate 10 μ m (0, 8 hour) and 20 μ m (72 hour). (B) IF staining of FAP-1 (green) and F-actin (red) in FAP-1^{+/+} and FAP-1^{-/-} HEL cells after PMA treatment for up to 8 hours. The scale bars indicate 20 μ m. (C) IF staining of F-actin (red) in PMA-treated HEL cells after treatment with the dimethyl sulfoxide (DMSO) control or 0.5 μ M jasplakinolide for 8 hours. The scale bars indicate 20 μ m. (D) Rescue experiment in FAP-1^{-/-} HEL cells transduced with Flag-tagged WT FAP-1 or the phosphatase-dead C2394S mutant. The cells were stained for Flag (green) and F-actin (red). White arrowheads indicate Flag-positive cells. The scale bar indicates 20 μ m. Images in panels B-D are representative of 3 independent experiments.

pre-DMS, driven by actin polymerization near GPIb-V-IX; and expansion into mature DMS via fusion with trans-Golgi-derived vesicles,³⁴⁻³⁸ requiring perinuclear actin disassembly. The presence of perinuclear pre-DMS structures in FAP-1^{ΔP/ΔP} MKs (Figure 3C-D; supplemental Figure 10C-D) suggests arrest at the pre-DMS to DMS transition, potentially reflecting defective actin depolymerization. To test this, the F-actin dynamics were examined in cultured MKs using anti-GPIb and tetramethylrhodamine isothiocyanate-phalloidin, which showed similar co-localization during invagination and pre-DMS formation (0 and 8 hours) in WT and FAP-1^{ΔP/ΔP} MKs (Figure 5A), but upon DMS expansion, F-actin dispersed in WT MKs but remained perinuclear in FAP-1^{ΔP/ΔP} MKs (Figure 5A; 72 hours), suggesting that FAP-1 facilitates the pre-DMS to DMS transition via actin depolymerization.

To examine this in human MKs, we generated a FAP-1 CRISPR-knockout HEL cell line and induced differentiation with phorbol 12-myristate 13-acetate (PMA), a commonly used model of MK differentiation.³⁹ At 2 hours after PMA addition, F-actin was predominantly perinuclear in both FAP-1^{+/+} and FAP-1^{-/-} cells,

whereas by 8 hours, it dispersed in the FAP-1^{+/+} cells but remained perinuclear in the FAP-1^{-/-} cells (Figure 5B), resembling cells co-treated with PMA and the actin-stabilizing agent jasplakinolide⁴⁰ (Figure 5C). Biochemical analyses showed that F-actin disassembly peaked at 4 hours in the FAP-1^{+/+} cells but was largely absent in the FAP-1^{-/-} cells, which produced a higher F-/G-actin ratio, confirming reduced actin depolymerization (supplemental Figure 12).

Notably, transient FAP-1 expression co-localized with perinuclear F-actin at 2 hours and disappeared by 8 hours in FAP-1^{+/+} HEL cells, coinciding with F-actin dispersal (Figure 5B) and suggesting that FAP-1 promotes perinuclear actin disassembly. Consistently, re-expression of WT, but not catalytically inactive FAP-1 (C2394S) restored cytoplasmic F-actin in FAP-1^{-/-} cells (Figure 5D), indicating that FAP-1 enzymatic activity contributes to actin depolymerization.

Together, these results suggest that FAP-1 facilitates perinuclear actin disassembly, thereby enabling DMS expansion, whereas its loss may arrest MKs at the pre-DMS stage.

FAP-1 deficiency causes sustained Src activation and cofilin inactivation, contributing to impaired DMS maturation

As documented, TPO initiates membrane invagination by inducing actin polymerization near GPIb-V-IX through CDC42 and neuronal Wiskott-Aldrich syndrome protein (N-WASP),^{35,36} potentially involving Src-mediated phosphorylation of CDC42(Y64) and WASP(Y291).^{35,41,42} Activated CDC42 then promotes p21-activated kinase–LIM kinase (PAK-LIMK)–dependent phosphorylation and inactivation of cofilin(S3),^{43,44} whereas subsequent perinuclear cofilin reactivation is required for DMS extension in MKs.^{35,45}

Because Src is an FAP-1 substrate,⁴⁶ we hypothesized that FAP-1 promotes DMS extension by inactivating Src, thereby enabling cofilin activation and actin depolymerization. Consistently, FAP-1^{+/+} cells exhibited transient perinuclear FAP-1 at 2 hours after PMA addition, colocalizing with F-actin, which decreased by 8 hours and was accompanied by reduced perinuclear pY416-Src (active) and pS3-cofilin (inactive) (Figure 6A–C). In contrast, FAP-1^{−/−} cells maintained perinuclear pY416-Src and pS3-cofilin from 2 to 8 hours, as confirmed by immunoblotting (Figure 6B–D).

Pharmacologic Src inhibition with dasatinib⁴⁷ dispersed perinuclear F-actin and restored the DMS architecture in FAP-1–deficient MKs and HEL cells (Figure 6E–F). In vivo, dasatinib treatment improved the MK abnormalities and attenuated the MF-like features in FAP-1^{ΔP/ΔP} mice with modest improvement in the hematologic parameters in asymptomatic animal (supplemental Figure 13). These findings identify FAP-1 as a key regulator of the pre-DMS to DMS transition via transient Src dephosphorylation, thereby enabling cofilin activation and F-actin depolymerization. Targeting Src with dasatinib may help to restore the DMS architecture and mitigate the MF phenotypes associated with an FAP-1 deficiency.

Given the link between JAK-STAT activation and MK hyperplasia in PMF, we also examined whether FAP-1 influences this pathway. FAP-1 knockdown enhanced ligand-dependent JAK-STAT signaling in M-07e cells (with WT JAK2), and increased pJAK2 and pSTAT5 levels were observed in both young and aged FAP-1^{ΔP/ΔP} marrow when compared with WT marrow with the highest levels measured in aged mice (supplemental Figure 14), which suggests that FAP-1 may modulate JAK-STAT signaling, thereby contributing to MK expansion in FAP-1^{ΔP/ΔP} mice.

Reduced FAP-1 expression and related pathogenic changes are associated with DMS abnormalities in patients with PMF

To assess the relevance of FAP-1 in human PMF, we examined the DMS architecture and FAP-1–Src signaling in bone marrow MKs from patients with PMF, ET, or lymphoma. The MK numbers were higher in ET and PMF when compared with lymphoma (Figure 7A). IF showed that >85% to 90% of MKs in ET and lymphoma had normal DMS, whereas 14 of the 16 patients with PMF had >25% pre-DMS MKs (26% to 95%), resembling FAP-1^{ΔP/ΔP} MKs, which occurred more frequently in PMF than in ET or lymphoma (Figure 7B–C; supplemental Table 3). Higher pre-DMS proportions were associated with lower platelet counts and increased fibrosis (Figure 7D–E; supplemental Table 3), suggesting a relationship between pre-DMS accumulation and advanced disease.

We subsequently analyzed FAP-1 expression and pY416-Src localization relative to the perinuclear DMS architecture (reflected by actin staining) in serial bone marrow sections. In lymphoma and ET, most MKs with mature DMS exhibited low or undetectable FAP-1 and minimal pY416-Src, whereas a minority of pre-DMS MKs showed detectable FAP-1 with weak or absent pY416-Src (Figure 7F–G). In contrast, PMF MKs showed frequent pre-DMS accumulation with reduced or absent FAP-1 and enhanced perinuclear pY416-Src (Figure 7F–G), which resembled FAP-1^{ΔP/ΔP} MKs and were absent in WT MKs (Figure 7G).

Double IF for FAP-1 and GPIb showed that reduced FAP-1 coincided with abnormal GPIb distribution in PMF but not in lymphoma or ET (supplemental Figure 15A). RNA in situ hybridization revealed lower FAP-1 transcripts in PMF MKs but preserved expression in ET and lymphoma, suggesting that reduced RNA contributes to decreased protein levels (supplemental Figure 15B). These findings are consistent with an association between reduced FAP-1, enhanced Src activation, and abnormal DMS organization in PMF.

Discussion

Although FAP-1 has been implicated in multiple signaling pathways,¹ its in vivo role remains unclear. In this study, FAP-1^{ΔP/ΔP} mice developed early and persistent MK abnormalities with platelet dysfunction from 2 to 4 months of age, preceding age-dependent MF-like changes in a subset of animals (~50%). Among the affected mice, most remained in a prefibrotic state, whereas only a fraction progressed to overt fibrosis, indicating highly penetrant MK defects with secondary and heterogeneous fibrotic progression, thereby supporting a primary role of FAP-1 in MK maturation.

Although FAP-1 has not been implicated in MF-like murine models previously, in vitro studies suggest a context- and cell type-dependent role in MK differentiation in K562 and HEL cells.^{48,49} In contrast, FAP-1^{ΔP/ΔP} mice support a physiological role for FAP-1 in MK maturation with relevance to MF-like features. Polyploidization and α -granule formation are preserved in asymptomatic mice, whereas symptomatic mice show a shift toward lower ploidy, consistent with the changes reported in MF.^{12,26} FAP-1 deficiency induces an age-independent, pre-DMS arrest with sustained Src activation, cofilin inactivation, and impaired F-actin disassembly. Given that actin remodeling during MK differentiation is tightly stage-regulated^{34–36} and that active Src inhibits cofilin-mediated actin severing,^{50,51} disruption of this axis leads to thrombocytopenia, MK accumulation, DMS abnormalities, and perinuclear F-actin aggregation.^{45,52} These features resemble the phenotype of FAP-1^{ΔP/ΔP} mice and support a model in which FAP-1 promotes cofilin-dependent F-actin disassembly through Src dephosphorylation during DMS maturation.

Functionally, this maturation defect translates into impaired thrombopoiesis and qualitative platelet dysfunction. In young FAP-1^{ΔP/ΔP} mice, partial DMS and proplatelet defects are already evident and are sufficient to maintain near-normal platelet counts, yet platelets exhibit prolonged bleeding times and defective clot retraction, indicating impaired hemostasis. Clinically, thrombosis predominates in polycythemia vera and ET, possibly reflecting the expansion of mature, proplatelet-forming MKs, whereas major bleeding is

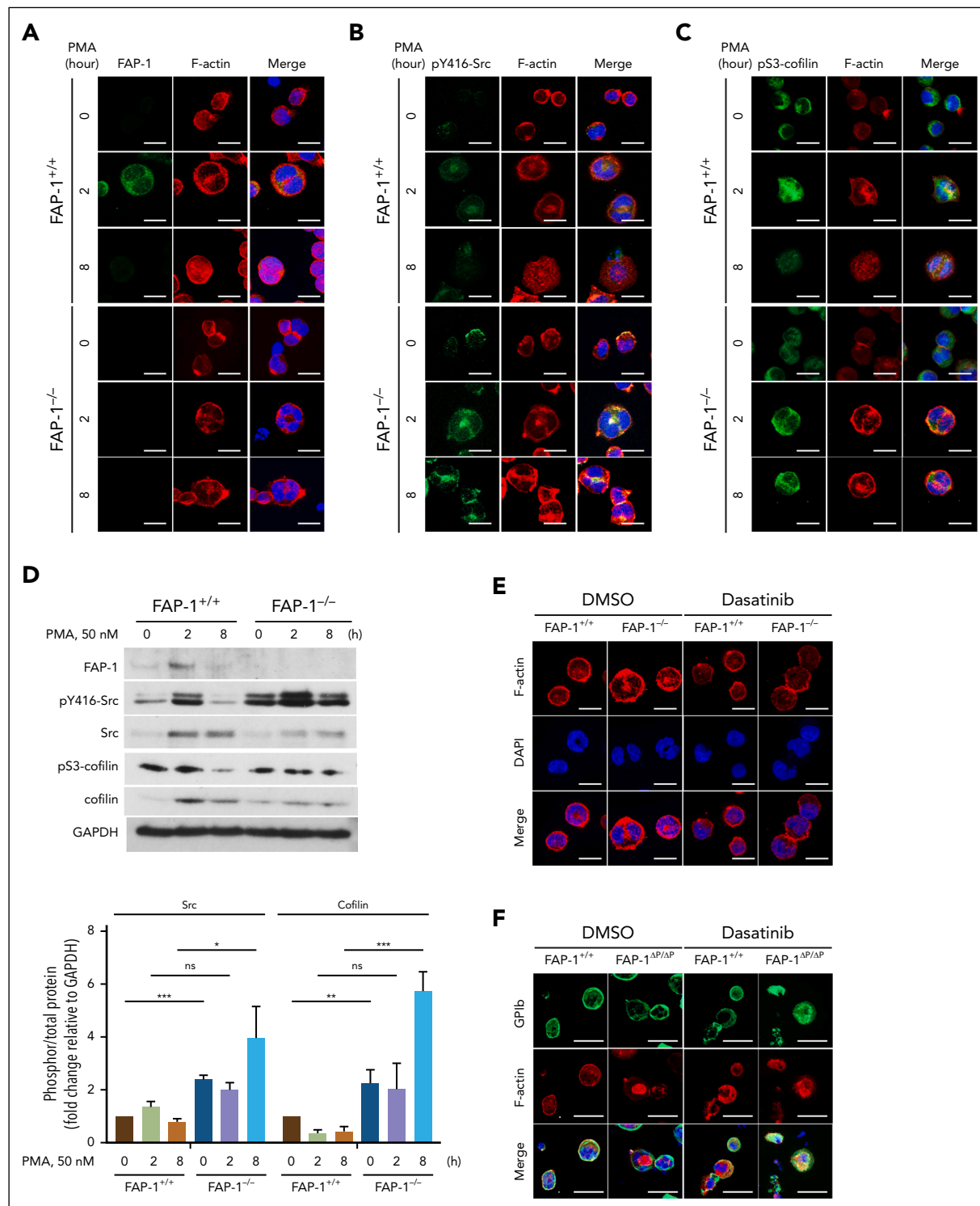


Figure 6. FAP-1 deficiency sustains Src activation and cofilin inactivation, impairing actin depolymerization and DMS maturation. (A-C) IF staining analysis of HEL cells stimulated with PMA for the indicated times. The cells were stained for F-actin (red) and either FAP-1 (A), pY416-Src (B), or pS3-cofilin (C) (green). The scale bars indicate 20 μ m. (D) Upper panel: western blot analysis of the indicated proteins in PMA-treated HEL cells. Lower panel: bar graphs show the quantification of phosphorylated to total protein ratios normalized to GAPDH (N = 3 independent experiments). The band intensities were quantified using ImageJ and are presented as the mean $\pm$ standard deviation. ns, not significant; * P < .05; ** P < .01; *** P < .001. (E) IF staining of F-actin (red) in PMA-treated HEL cells after treatment with the DMSO control or 10 μ M Src inhibitor dasatinib for 8 hours. The scale bars indicate 20 μ m. (F) IF staining of GPIb (green) and F-actin (red) in primary MKs treated with DMSO or 10 μ M dasatinib for 24 hours. The scale bars indicate 50 μ m. Images in panels A-C and E-F are representative of at least 3 independent experiments. GAPDH, glyceraldehyde-3-phosphate dehydrogenase.

more frequent in PMF, potentially owing to MK dysplasia and platelet dysfunction, and increases from the prefibrotic to overt stage.^{12,13,53-56} Consistent with this clinical pattern, FAP-1 ^{Δ P/ Δ P}

mice exhibited platelet dysfunction, accompanied by spontaneous intra-abdominal hemorrhage, supporting a bleeding-prone rather than a thrombosis-dominant course.

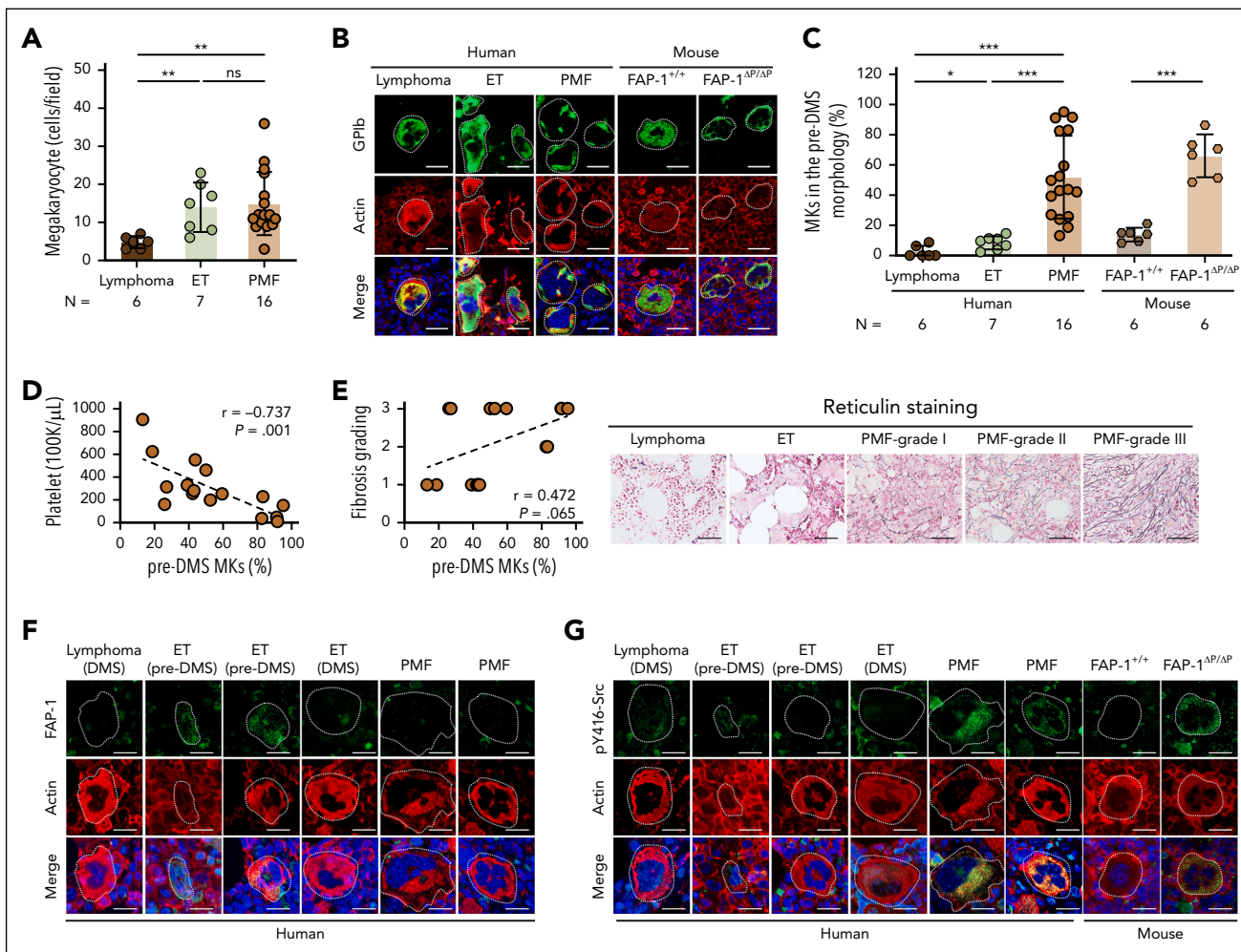


Figure 7. Downregulation of FAP-1 and sustained Src activation correlate with DMS abnormalities in patients with PMF. (A) Quantification of the MK density (cells per field) in bone marrow biopsies from patients with lymphoma (N = 6), ET (N = 7), and PMF (N = 16). (B) Representative IF staining of GPIb (green) and actin (red) in human and mouse bone marrow sections. The nuclei were counterstained with DAPI (4',6-diamidino-2-phenylindole; blue). The scale bar indicates 20 μ m. (C) Quantification of the percentage of MKs in the pre-DMS morphology in the indicated human and mouse cohorts. The percentage of MKs in the pre-DMS morphology for each patient is listed in supplemental Table 3. (D) The correlation analysis between the percentage of pre-DMS MKs and platelet counts in patients with PMF (N = 16). The Pearson correlation coefficient (r) and P value are indicated. (E) Left panel: correlation analysis between the percentage of pre-DMS MKs and fibrosis grade in patients with PMF. Right panel: representative images of reticulin staining are shown. The scale bars indicate 50 μ m. (F-G) Representative IF staining images of FAP-1 (F) and pY416-Src (G) in bone marrow samples. The white dotted lines in each image delineate MK. The scale bars indicate 20 μ m. Data in panels A and C are presented as mean $\pm$ standard deviation. Statistical significance was determined using a t test. * P < .05, ** P < .01, *** P < .001; ns, not significant.

In line with these observations, the phenotypic spectrum of FAP-1 $\Delta P/\Delta P$ mice is broadly consistent with select features of human PMF in which the progression to advanced fibrosis is limited to a subset of patients^{15,16} and EMH-associated hepatosplenomegaly can occur in prefibrotic disease,^{17,18} reflecting a partial dissociation between clinical manifestations and fibrosis stage. Consistently, MK DMS abnormalities across the fibrosis stages²⁴ suggest that FAP-1-dependent MK maturation is an early event relevant to MF-associated MK dysfunction. Notably, the reduced FAP-1 expression, abnormal DMS and actin organization, and increased Src activation observed in PMF MKs suggest that FAP-1-dependent pathways may contribute to MK and platelet abnormalities in PMF. Given the impaired clot retraction in FAP-1-deficient mice, similar clot retraction defects may also occur in patients with PMF, a possibility that warrants further investigation.

Despite the impaired platelet function, platelet counts remain relatively preserved in young FAP-1 $\Delta P/\Delta P$ mice, raising the

question of how thrombopoiesis is maintained. One explanation is that impaired platelet function reduces platelet-mediated TPO clearance, thereby increasing TPO-JAK/STAT signaling that may support MK proliferation and platelet maintenance.^{57,58} Consistent with this, FAP-1 deficiency increases JAK/STAT activity in vitro and in vivo. Although the mechanism that links FAP-1 deficiency to JAK2 activation, potentially involving Src activation, remains unclear, these observations suggest that altered TPO homeostasis and enhanced JAK/STAT signaling may drive early MK hyperplasia in young FAP-1 $\Delta P/\Delta P$ mice, thus supporting a role for FAP-1 as a negative regulator of cytokine-driven signaling. A loss of FAP-1 may also increase cytokine sensitivity, thereby potentially contributing to trilineage hyperplasia and organomegaly via amplified JAK/STAT signaling downstream of multiple cytokine receptors.⁵⁹ With disease progression in aged mice, this early hyperproliferative, yet dysfunctional, state seems to shift toward ineffective thrombopoiesis with severe DMS disorganization and markedly reduced proplatelet formation,

consistent with thrombocytopenia despite persistent MK hyperplasia.

To determine whether these progressive changes are hematopoietic cell-intrinsic, we performed transplantation experiments. Marrow from symptomatic, but not young or asymptomatic, FAP-1^{ΔP/ΔP} donors transferred the DMS defects, thrombocytopenia, and MF-like features to WT recipients. Furthermore, the FAP-1^{ΔP/ΔP} mice exhibited a spectrum of fibrosis severity with increased marrow cellularity in the prefibrotic stages and reduced cellularity in advanced fibrosis, suggesting progression toward marrow failure. These stage-dependent differences in disease severity likely underlie the preserved repopulation capacity observed in transplantation experiments, and future studies using advanced (MF-3) donors will be required to define the impact of severe fibrosis.

Taken together, these findings support a model in which MPL/JAK2 signaling creates a pro-megakaryopoietic state, whereas age-dependent intrinsic defects in mutant hematopoietic cells impair MK maturation, thrombopoiesis, and contribute to fibrotic changes. Aging-associated factors may further exacerbate DMS and proplatelet defects within this disease model. Impaired DMS maturation in FAP-1-deficient MKs is associated with defective proplatelet formation and platelet dysfunction, whereas chronic accumulation of immature, cytokine-producing MKs may contribute to fibrosis and EMH during aging, consistent with sequential manifestations that arise from a primary MK maturation defect.

Importantly, the fibrotic changes observed in the FAP-1^{ΔP/ΔP} mice are unlikely to represent purely reactive or secondary fibrosis, which is typically associated with inflammatory, autoimmune, infectious, irradiation, or toxic insults and is generally not accompanied by intrinsic MK atypia.^{60,61} In contrast, FAP-1^{ΔP/ΔP} mice exhibited MK hyperplasia, cloud-like nuclear morphology, and defective DMS maturation, indicating a primary defect in the MK biology. These findings, together with evidence for a hematopoietic cell-intrinsic mechanism, support a model in which intrinsic MK dysfunction contributes to MF progression rather than reactive fibrosis.

From a translational perspective, human MF is typically diagnosed after prefibrotic or overt disease is established, limiting access to early MK maturation defects and platelet dysfunction. In contrast, FAP-1^{ΔP/ΔP} mice display early DMS abnormalities, impaired proplatelet formation, and platelet dysfunction that precede overt disease. These findings suggest that early MK and platelet abnormalities may serve as indicators of PMF-like evolution and that targeting MK cytoskeletal regulation could influence disease progression, a hypothesis that requires further validation.

To address this, we focused on Src, a proto-oncogenic tyrosine kinase and FAP-1 substrate,⁴⁶ which is upregulated during MK maturation. Constitutive Src activation has been associated with thrombocytopenia and MF in both human and murine models with murine studies showing no overt polyploidization defects.⁶²⁻⁶⁴ Our data demonstrated that a FAP-1 deficiency sustained Src activation and impaired MK maturation. Pharmacologic Src inhibition restored the DMS architecture and resolved aberrant perinuclear F-actin accumulation in vitro and partially ameliorated the MF-like features in FAP-1^{ΔP/ΔP} mice and modestly improved

platelet counts in asymptomatic mice, raising the possibility that early-stage targeting of the FAP-1/Src axis may help to mitigate the bleeding complications associated with platelet dysfunction or delay the progression to overt MF, although longer-term studies are needed to evaluate these effects.

RNA in situ hybridization further indicated reduced FAP-1 transcripts in PMF MKs, suggesting possible transcriptional regulation (eg, promoter methylation¹). Given that most analyzed ET and PMF cases harbor JAK2 mutations, reduced FAP-1 may cooperate with these driver mutations in disease development. Collectively, these data link a reduction in FAP-1 to defective DMS organization in PMF and support further mechanistic investigation, including evaluations of Src inhibition alone or in combination with JAK/STAT inhibitors or standard therapies, such as hydroxyurea⁶⁵ or ropeginterferon alfa-2b,⁶⁶ while causality remains to be established.

The actin cytoskeleton undergoes continuous ATP-dependent treadmilling, which is accelerated by cofilin-mediated filament severing to replenish monomers and enable rapid remodeling.⁶⁷ Notably, Src activation functions as a critical switch that inhibits cofilin-mediated actin severing.^{50,51} In MKs, our data identified FAP-1 as a critical regulator of Src-dependent, cofilin-mediated actin severing during DMS maturation. Given that rapid actin remodeling underlies diverse processes, including axon guidance,⁶⁸ synapse development,⁶⁹ and cytokinesis,⁷⁰ further defining FAP-1 regulation may provide broader insight into disease pathogenesis and potential therapeutic opportunities.

In conclusion, our study identified a previously unrecognized role for FAP-1 in regulating Src-cofilin-dependent actin remodeling during DMS maturation in MKs. FAP-1 loss impairs MK maturation and thrombopoiesis and is associated with age-dependent fibrotic remodeling, highlighting the contribution of MK maturation defects to disease evolution and implicating the FAP-1/Src axis as a potential therapeutic target with possible relevance to bleeding and fibrotic progression in PMF.

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Authorship

Contribution: M.-F.C., K.-H.Y., P.-J.C., and S.-H.Y. designed the research; M.-F.C. performed the experiments; K.-P.C. provided the clinical formalin-fixed paraffin-embedded samples and performed the clinical reticulin grading; K.-P.C. and C.-W.L. analyzed and characterized the

pathologic tissue sections; C.-C.L. characterized the blood cell types in blood smears; C.-C.L. and W.-C.C. provided the complete blood count data of clinical samples; M.-F.C. and S.-H.Y. drafted the manuscript; and K.-H.Y., P.-J.C., and W.-C.C. critically revised the manuscript.

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Footnotes

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The publication-related data are available from the corresponding author upon reasonable request.

The online version of this article contains a data supplement.

There is a *Blood Commentary* on this article in this issue.

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REFERENCES

1. McHeik S, Aptekar L, Coopman P, D’Hondt V, Freiss G. Dual role of the PTPN13 tyrosine phosphatase in cancer. *Biomolecules*. 2020;10(12):1659.

2. Yeh SH, Wu DC, Tsai CY, et al. Genetic characterization of fas-associated phosphatase-1 as a putative tumor suppressor gene on chromosome 4q21.3 in hepatocellular carcinoma. *Clin Cancer Res*. 2006;12(4):1097-1108.

3. Huang W, Bei L, Eklund EA. Fas-associated phosphatase 1 (Fap1) influences betacatenin activity in myeloid progenitor cells expressing the Bcr-abl oncogene. *J Biol Chem*. 2013;288(18):12766-12776.

4. Wansink DG, Peters W, Schaafsma I, et al. Mild impairment of motor nerve repair in mice lacking PTP-BL tyrosine phosphatase activity. *Physiol Genomics*. 2004;19(1):50-60.

5. Lorber B, Hendriks WJ, Van der Zee CE, Berry M, Logan A. Effects of LAR and PTP-BL phosphatase deficiency on adult mouse retinal cells activated by lens injury. *Eur J Neurosci*. 2005;21(9):2375-2383.

6. Nakahira M, Tanaka T, Robson BE, Mizgerd JP, Grusby MJ. Regulation of signal transducer and activator of transcription signaling by the tyrosine phosphatase PTP-BL. *Immunity*. 2007;26(2):163-176.

7. Lee HJ, Daver N, Kantarjian HM, Verstovsek S, Ravandi F. The role of JAK pathway dysregulation in the pathogenesis and treatment of acute myeloid leukemia. *Clin Cancer Res*. 2013;19(2):327-335.

8. Spivak JL. Myeloproliferative Neoplasms. *N Engl J Med*. 2017;376(22):2168-2181.

9. Vainchenker W, Kralovics R. Genetic basis and molecular pathophysiology of classical myeloproliferative neoplasms. *Blood*. 2017;129(6):667-679.

10. Babakhanlou R, Masarova L, Verstovsek S. A review of essential thrombocythemia and its complications. *Clin Adv Hematol Oncol*. 2023;21(2):76-84.

11. Thiele J, Kvasnicka HM, Mullauer L, Buxhofer-Ausch V, Gisslinger B, Gisslinger H. Essential thrombocythemia versus early primary myelofibrosis: a multicenter study to validate the WHO classification. *Blood*. 2011;117(21):5710-5718.

12. Balduini A, Badalucco S, Pugliano MT, et al. In vitro megakaryocyte differentiation and proplatelet formation in Ph-negative classical myeloproliferative neoplasms: distinct patterns in the different clinical phenotypes. *PLoS One*. 2011;6:e21015.

13. Kc D, Falchi L, Verstovsek S. The underappreciated risk of thrombosis and bleeding in patients with myelofibrosis: a review. *Ann Hematol*. 2017;96(10):1595-1604.

14. Malara A, Abbonante V, Zingariello M, Migliaccio A, Balduini A. Megakaryocyte contribution to bone marrow fibrosis: many arrows in the quiver. *Mediterr J Hematol Infect Dis*. 2018;10(1):e2018068.

15. Jeryczynski G, Thiele J, Gisslinger B, et al. Pre-fibrotic/early primary myelofibrosis vs. WHO-defined essential thrombocythemia: the impact of minor clinical diagnostic criteria on the outcome of the disease. *Am J Hematol*. 2017;92(9):885-891.

16. Carobbio A, Guglielmelli P, Rumi E, et al. A multistate model of survival prediction and event monitoring in prefibrotic myelofibrosis. *Blood Cancer J*. 2020;10(10):100.

17. Guglielmelli P, Rotunno G, Pacilli A, et al. Differences in clinical and molecular characteristics and outcome in prefibrotic and overt primary myelofibrosis according to 2016 WHO criteria: a study on 639 patients of the Agimm Group. *Blood*. 2016;128(22):943.

18. Cenariu D, Iluta S, Zimta AA, et al. Extramedullary hematopoiesis of the liver and spleen. *J Clin Med*. 2021;10(24):5831.

19. Jung SH, Lee SE, Yun S, et al. Different inflammatory, fibrotic, and immunological signatures between pre-fibrotic and overt primary myelofibrosis. *Haematologica*. 2025;110(4):938-951.

20. Tefferi A. Primary myelofibrosis: 2023 update on diagnosis, risk-stratification, and management. *Am J Hematol*. 2023;98(5):801-821.

21. Du C, Chen J, Wang J. New insights into the generation and function of megakaryocytes in health and disease. *Haematologica*. 2025;110(7):1500-1512.

22. Deutsch VR, Tomer A. Advances in megakaryocytopoiesis and thrombopoiesis: from bench to bedside. *Br J Haematol*. 2013;161(6):778-793.

23. Ivanovna Gabrilchak A, Anatolieva Gusyakova O, Aleksandrovich Antipov V, Alekseevna Medvedeva E, Leonidovna Tukshumskaya L. A modern overview of the process of platelet formation (thrombocytopoiesis) and its dependence on several factors. *Biochem Med (Zagreb)*. 2024;34(3):030503.

24. Zingariello M, Rosti V, Vannucchi AM, et al. Shared and distinctive ultrastructural abnormalities expressed by megakaryocytes in bone marrow and spleen from patients with myelofibrosis. *Front Oncol*. 2020;10:584541.

25. Hattori A, Koike K, Ito S, Matsuoka M. Static and functional morphology of the pathological platelets in primary myelofibrosis and myeloproliferative syndrome. *Ser Haematol*. 1975;8(1):126-150.

26. Melo-Cardenas J, Migliaccio AR, Crispino JD. The role of megakaryocytes in myelofibrosis. *Hematol Oncol Clin North Am*. 2021;35(2):191-203.

27. Lin CC, Yao CY, Wang YH, et al. IDH2 mutation accelerates TPO-induced myelofibrosis with enhanced S100a8/a9 and NFkappaB signaling in vivo. *EJHaem*. 2024;5(4):738-748.

28. Strassel C, Eckly A, Leon C, et al. Hirudin and heparin enable efficient megakaryocyte differentiation of mouse bone marrow progenitors. *Exp Cell Res*. 2012;318(1):25-32.

29. Liu Y, Jennings NL, Dart AM, Du XJ. Standardizing a simpler, more sensitive and accurate tail bleeding assay in mice. *World J Exp Med*. 2012;2(2):30-36.
30. Vannucchi AM, Bianchi L, Cellai C, et al. Development of myelofibrosis in mice genetically impaired for GATA-1 expression (GATA-1(low) mice). *Blood*. 2002;100(4):1123-1132.
31. Alexander WS. Thrombopoietin and the c-Mpl receptor: insights from gene targeting. *Int J Biochem Cell Biol*. 1999;31(10):1027-1035.
32. Peck-Radosavljevic M. Thrombocytopenia in chronic liver disease. *Liver Int*. 2017;37(6):778-793.
33. Agarwal A, Morrone K, Bartenstein M, Zhao ZJ, Verma A, Goel S. Bone marrow fibrosis in primary myelofibrosis: pathogenic mechanisms and the role of TGF-beta. *Stem Cell Investig*. 2016;3:5.
34. Eckly A, Heijnen H, Pertuy F, et al. Biogenesis of the demarcation membrane system (DMS) in megakaryocytes. *Blood*. 2014;123(6):921-930.
35. Antkowiak A, Viaud J, Severin S, et al. Cdc42-dependent F-actin dynamics drive structuration of the demarcation membrane system in megakaryocytes. *J Thromb Haemost*. 2016;14(6):1268-1284.
36. Ghalloussi D, Dhenge A, Bergmeier W. New insights into cytoskeletal remodeling during platelet production. *J Thromb Haemost*. 2019;17(9):1430-1439.
37. Chen Q, Pollard TD. Actin filament severing by cofilin dismantles actin patches and produces mother filaments for new patches. *Curr Biol*. 2013;23(13):1154-1162.
38. Eitzen G. Actin remodeling to facilitate membrane fusion. *Biochim Biophys Acta*. 2003;1641(2-3):175-181.
39. Long MW, Heffner CH, Williams JL, Peters C, Prochownik EV. Regulation of megakaryocyte phenotype in human erythroleukemia cells. *J Clin Invest*. 1990;85(4):1072-1084.
40. Holzinger A. Jaspaklinolide: an actin-specific reagent that promotes actin polymerization. *Methods Mol Biol*. 2009;586:71-87.
41. Tu S, Wu WJ, Wang J, Cerione RA. Epidermal growth factor-dependent regulation of Cdc42 is mediated by the Src tyrosine kinase. *J Biol Chem*. 2003;278(49):49293-49300.
42. Bussell K. Another two stings in WASP's tail. *Nat Rev Mol Cell Biol*. 2003;4(7):510.
43. Melendez J, Grogg M, Zheng Y. Signaling role of Cdc42 in regulating mammalian physiology. *J Biol Chem*. 2011;286(4):2375-2381.
44. Wioland H, Guichard B, Senju Y, et al. ADF/Cofilin accelerates actin dynamics by severing filaments and promoting their depolymerization at both ends. *Curr Biol*. 2017;27(13):1956-1967.e7.
45. Bender M, Eckly A, Hartwig JH, et al. ADF/cofilin-dependent actin turnover determines platelet formation and sizing. *Blood*. 2010;116(10):1767-1775.
46. Wang J, Li S, Zhang X, et al. Protein tyrosine phosphatase PTPL1 suppresses lung cancer through Src/ERK/YAP1 signaling. *Thorac Cancer*. 2022;13(21):3042-3051.
47. Araujo J, Logothetis C. Dasatinib: a potent SRC inhibitor in clinical development for the treatment of solid tumors. *Cancer Treat Rev*. 2010;36(6):492-500.
48. Sardina JL, Lopez-Ruano G, Prieto-Bermejo R, et al. PTPN13 regulates cellular signalling and β -catenin function during megakaryocytic differentiation. *Biochim Biophys Acta*. 2014;1843(12):2886-2899.
49. Moshiri H, Cabrera Riofrio DA, Lim YJ, et al. Germline PTPN13 mutations in patients with bone marrow failure and acute lymphoblastic leukemia. *Leukemia*. 2022;36(8):2132-2135.
50. Tehrani S, Tomasevic N, Weed S, Sakowicz R, Cooper JA. Src phosphorylation of cortactin enhances actin assembly. *Proc Natl Acad Sci U S A*. 2007;104(29):11933-11938.
51. Yoo Y, Ho HJ, Wang C, Guan JL. Tyrosine phosphorylation of cofilin at Y68 by v-Src leads to its degradation through ubiquitin-proteasome pathway. *Oncogene*. 2010;29(2):263-272.
52. Kile BT, Panopoulos AD, Stirzaker RA, et al. Mutations in the cofilin partner Aip1/Wdr1 cause autoinflammatory disease and macrothrombocytopenia. *Blood*. 2007;110(7):2371-2380.
53. Finazzi G, Carobbio A, Thiele J, et al. Incidence and risk factors for bleeding in 1104 patients with essential thrombocythemia or prefibrotic myelofibrosis diagnosed according to the 2008 WHO criteria. *Leukemia*. 2012;26(4):716-719.
54. Enblom-Larsson A, Renlund H, Andreasson B, Holmberg H, Liljeholm M, Sjalander A. Thromboembolic events, major bleeding and mortality in essential thrombocythaemia and polycythaemia vera-A matched nationwide population-based study. *Br J Haematol*. 2024;204(5):1740-1751.
55. De Stefano V, Za T, Rossi E, et al. Recurrent thrombosis in patients with polycythemia vera and essential thrombocythemia: incidence, risk factors, and effect of treatments. *Haematologica*. 2008;93(3):372-380.
56. Michiels JJ, Berneman Z, Schroyens W, Finazzi G, Budde U, van Vliet HH. The paradox of platelet activation and impaired function: platelet-von Willebrand factor interactions, and the etiology of thrombotic and hemorrhagic manifestations in essential thrombocythemia and polycythemia vera. *Semin Thromb Hemost*. 2006;32(6):589-604.
57. Ng AP, Kauppi M, Metcalf D, et al. Mpl expression on megakaryocytes and platelets is dispensable for thrombopoiesis but essential to prevent myeloproliferation. *Proc Natl Acad Sci U S A*. 2014;111(16):5884-5889.
58. Makar RS, Zhukov OS, Sahud MA, Kuter DJ. Thrombopoietin levels in patients with disorders of platelet production: diagnostic potential and utility in predicting response to TPO receptor agonists. *Am J Hematol*. 2013;88(12):1041-1044.
59. Woods B, Chen W, Chiu S, et al. Activation of JAK/STAT signaling in megakaryocytes sustains myeloproliferation in vivo. *Clin Cancer Res*. 2019;25(19):5901-5912.
60. Gianelli U, Cattaneo D, Bossi A, et al. The myeloproliferative neoplasms, unclassifiable: clinical and pathological considerations. *Mod Pathol*. 2017;30(2):169-179.
61. Ghosh K, Shome DK, Kulkarni B, Ghosh MK, Ghosh K. Fibrosis and bone marrow: understanding causation and pathobiology. *J Transl Med*. 2023;21(1):703.
62. Turro E, Greene D, Wijgaerts A, et al. A dominant gain-of-function mutation in universal tyrosine kinase SRC causes thrombocytopenia, myelofibrosis, bleeding, and bone pathologies. *Sci Transl Med*. 2016;8(328):328ra30.
63. Barozzi S, Di Buduo CA, Marconi C, et al. Pathogenetic and clinical study of a patient with thrombocytopenia due to the p.E527K gain-of-function variant of SRC. *Haematologica*. 2021;106(3):918-922.
64. Nagy Z, Mori J, Ivanova VS, Mazharian A, Senis YA. Interplay between the tyrosine kinases Chk and Csk and phosphatase PTPRJ is critical for regulating platelets in mice. *Blood*. 2020;135(18):1574-1587.
65. Grunwald MR, Ritchie EK, Rumi E, et al. Treatment comparison of hydroxyurea versus ruxolitinib in essential thrombocythaemia: a matched-cohort analysis. *EJHaem*. 2024;5(4):778-783.
66. Kiladjian JJ, Marin FF, Al-Ali HK, et al. ROP-ET: a prospective phase III trial investigating the efficacy and safety of ropeginterferon alfa-2b in essential thrombocythemia patients with limited treatment options. *Ann Hematol*. 2024;103(7):2299-2310.
67. Goode BL, Eskin J, Shekhar S. Mechanisms of actin disassembly and turnover. *J Cell Biol*. 2023;222(12):e202309021.
68. Dent EW, Gupton SL, Gertler FB. The growth cone cytoskeleton in axon outgrowth and guidance. *Cold Spring Harb Perspect Biol*. 2011;3:a001800.
69. Zhang W, Benson DL. Stages of synapse development defined by dependence on F-actin. *J Neurosci*. 2001;21(14):5169-5181.
70. McDargh Z, Zhu T, Zhu H, O'Shaughnessy B. Actin turnover protects the cytokinetic contractile ring from structural instability. *J Cell Sci*. 2023;136(5):jcs259969.